

Vanishing of the original proper-source kernel through conductor coprimality

Split-Zero programme: native continuation

19 September 2026

Abstract

The supplied native continuation proves that the actual proper-source kernel is annihilated by the fourth power of the product of the four Fourier coordinates. We evaluate the remaining coprimality question in the original period family. For every retained arithmetic unit phase, the limiting conductor and coordinate divisor are coprime on the domain $\gamma \geq 7\delta > 0$. An explicit family of sixteen 10 by 6 coefficient matrices extends this to a finite large-period domain and to the complement of a finite exceptional set in the admitted period domain. Consequently the original proper-source kernel is zero there, at every original degree $k \geq 9$. The source Schur term disappears by an equality of the original spaces. The full target minimum remains in the metric receiving this result. All coordinates and unit factors are restored by explicit maps.

1 Original objects and the question settled here

Let $0 < \delta < 1/2$, $\gamma > 2$, and retain the ordered simple quartet

$$w_1 = \delta + i\gamma, \quad w_2 = \delta - i\gamma, \quad w_3 = -\delta + i\gamma, \quad w_4 = -\delta - i\gamma. \quad (\text{NC1})$$

Put $\beta = 2(\gamma^2 - \delta^2)$ and $\eta = (\delta^2 + \gamma^2)^2$. The centered polynomial is $h_c(w) = w^4 + \beta w^2 + \eta$. The evaluation matrix V has rows $(1, w_j, w_j^2, w_j^3)$. Retain the complete unit matrix $U = \text{diag}(a, \bar{a}, \bar{a}, a)$, where the actual $a \neq 0$ is fixed. The original period family from PCL1–7 [2] is

$$H(u) = D_u \mathcal{R}(u^{-1}) V^{-1} U, \quad D_u = e^{a/u} G_u, \quad \mathfrak{a} = 1/160 + \beta/24 + \eta/2. \quad (\text{NC2})$$

Here G_u is the original invertible diagonal matrix of Gamma ray constants and branch powers. Its values are retained. The entire matrix $\mathcal{R}$ has determinant one and

$$J_\infty = \mathcal{R}(0) = \begin{pmatrix} 1 & 0 & -\beta/3 & 0 \\ 0 & 1 & 0 & -2\beta/3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (\text{NC3})$$

For completeness, the full coefficients used below are

$$R_{br,n} = \sum_{\substack{p,h \geq 0 \\ 2p+4h=5n+r+1-b}} \frac{\beta^p \eta^h}{3^p p! h!} (-1)^{p+h-n} 5^{p+h-n} (b/5)_{p+h-n}, \quad 1 \leq b \leq 4, \quad 0 \leq r \leq 3. \quad (\text{NC4})$$

The convergence and determinant identity are the supplied PCL proof; they are not inferred from a finite truncation. This paper proves the additional coprimality and kernel conclusions from that exact family.

Write $D = \text{diag}(\zeta, \zeta^2, \zeta^3, \zeta^4)$, $\zeta = e^{2\pi i/5}$, and $gf(X) = f(D^{-1}X)$. In the literal Fourier coordinates $X = (X_1, \dots, X_4)$ set

$$Q_u(X) = Q_0(H(u)^{-1}X), \quad Q_0(t) = t_1t_4 - t_2t_3, \quad Q_{u,j} = g^j Q_u, \quad A_u = \prod_{j=1}^4 Q_{u,j}. \quad (\text{NC5})$$

Work on the original five-orbit domain: these five quadrics are nonassociate prime polynomials. Retain the original degrees $k \equiv 1 \pmod{4}$, $k \geq 9$. Let $R_u = \mathbb{C}[X]/(Q_u)$, $B_u = \text{im}(\mathbb{C}[X]^{(g)} \rightarrow R_u)$ and $C_{u,k} = (R_u/(A_u))_k$. Let $J_{u,k}$ denote the image of invariant degree- k polynomials in $C_{u,k}$; its original first- v -jet-vanishing subspace is denoted by $J_{v,k}$. The proper-source kernel under study is exactly

$$E_k = \{f \in J_{v,k} : X_i^4 f \in J_{u,k+4} \text{ for } i = 1, 2, 3, 4\}. \quad (\text{NC6})$$

All memberships here are in the specified coefficient quotients. These are the cofactor kernel and four multipliers of the supplied note E1–4 [1]. There is no replacement of this kernel by an arithmetic eigenspace or by the one-step observation kernel.

2 Return from the conductor to the actual source kernel

We give the algebraic argument, including the vertex, so that the kernel conclusion does not depend on a rank analogy.

Lemma 1. *The conductor $\{c \in R_u : cR_u \subset B_u\}$ equals $A_u R_u$. With $p = X_1 X_2 X_3 X_4$, the class $p^4 f$ vanishes in $C_{u,k+16}$ for every $f \in E_k$.*

Proof. For any homogeneous lift h , the invariant sum $\sum_{j=0}^4 g^j(A_u h)$ restricts on $Q_u = 0$ to $A_u h$: each other summand contains Q_u . Thus $A_u R_u \subset B_u$, without dividing the sum by five.

For the reverse inclusion localize at X_i^5 . The cyclic action on $X_i \neq 0$ is free. The localized polynomial ring is free over its invariants with basis $1, X_i, \dots, X_i^4$: each character can be cleared by an invertible power of X_i . The relation is $T^5 - X_i^5$ and its derivative is a unit. This faithfully flat base change turns the invariant-image inclusion into

$$\mathbb{C}[X]/\left(\prod_{j=0}^4 Q_{u,j}\right) \longrightarrow \bigoplus_{j=0}^4 \mathbb{C}[X]/(Q_{u,j}) \quad (\text{NC7})$$

on that chart. Indeed an invariant polynomial vanishing modulo Q_u vanishes modulo every $Q_{u,j}$, hence is divisible by their product; its quotient is again invariant. The free cyclic orbit gives (NC7). A tuple supported on its j th component lies in the image exactly when that component is divisible by all four other primes. Their ideals intersect in their product in the polynomial UFD. Every ideal in the direct sum decomposes componentwise, proving the conductor description on each $X_i \neq 0$ chart. Faithful flatness descends that description.

The smooth projective quadric has its original Segre coordinates, with $(R_u)_d = T_{d,d}$, the biforms of bidegree (d, d) . If a homogeneous c of degree d is in the conductor, c/A_u is a regular section of $\mathcal{O}(d-8, d-8)$ on all four X_i charts. They cover the projective quadric. On the standard Segre charts the regularity conditions say precisely that this section is a biform of that bidegree, or is zero if $d < 8$. Thus $c \in A_u R_u$ globally. There is no extra vertex-supported contribution.

Lift $f \in E_k$ to B_u . By (NC6) and $A_u R_u \subset B_u$, each $X_i^4 f$ is in B_u . For a monomial m of nonzero character w , choose i with $4i = w \pmod{5}$. The polynomial $p^4 m/X_i^4$ has nonnegative exponents and character zero. Hence $p^4 m f \in B_u$. For a weight-zero monomial use $f \in B_u$ directly. Monomials span R_u , so $p^4 f$ is in the conductor and therefore in $A_u R_u$. $\square$

In the original Segre substitution let $\ell_i = X_i|_{Q_u=0}$ and $\mathcal{A} = A_u|_{Q_u=0}$. They have bidegrees $(1, 1)$ and $(8, 8)$. The previous lemma and polynomial divisibility give the exact ambient multiplication kernel, with all multiplicities retained:

$$\ker M_{p^4} = (\mathcal{A}/G)T_{k-8+g_1, k-8+g_2}/\mathcal{A}T_{k-8, k-8}, \quad G = \gcd(\mathcal{A}, (\ell_1\ell_2\ell_3\ell_4)^4), \quad \text{bideg } G = (g_1, g_2). \quad (\text{NC8})$$

Indeed $\mathcal{A} \mid p^4 f$ is equivalent to $\mathcal{A}/G \mid f$ in the biform UFD. Consequently $G = 1$ implies $E_k = 0$. We now prove $G = 1$ on an explicit original period domain.

3 An exact auxiliary coordinate map, with its scalar retained

Set $z = u^{-1}$ and $Y = D_u^{-1}X$. This map is invertible, commutes with D , and carries every coordinate hyperplane to itself. Its exact polynomial identities are

$$\begin{aligned} Q_u(D_u Y) &= \tilde{Q}_z(Y) := Q_0(U^{-1}V\mathcal{R}(z)^{-1}Y), \\ A_u(D_u Y) &= \prod_{j=1}^4 \tilde{Q}_z(D^{-j}Y), \quad p(D_u Y) = \det(D_u)p(Y). \end{aligned} \quad (\text{NC9})$$

Thus coprimality and ideal membership are transported in both directions. This is a coordinate isomorphism for a divisibility calculation; all D_u factors stay in the original Hermitian metrics.

Write

$$t = \frac{\beta}{6} = \frac{\gamma^2 - \delta^2}{3} > 0, \quad d = 2\delta\gamma > 0, \quad \alpha = (\delta + i\gamma)^2 = -3t + id, \quad c = a^{-2} \neq 0. \quad (\text{NC10})$$

At $z = 0$, (NC3) gives the coefficient polynomial

$$P_Y(w) = (Y_1 + 2tY_3) + (Y_2 + 4tY_4)w + Y_3w^2 + Y_4w^3. \quad (\text{NC11})$$

The complete original quadric is consequently

$$\tilde{Q}_0(Y) = c\{[Y_1 + (\alpha + 2t)Y_3]^2 - \alpha[Y_2 + (\alpha + 4t)Y_4]^2\} - \bar{c}\{[Y_1 + (\bar{\alpha} + 2t)Y_3]^2 - \bar{\alpha}[Y_2 + (\bar{\alpha} + 4t)Y_4]^2\}. \quad (\text{NC12})$$

The subtraction is coefficientwise; it does not conjugate the variables. Introduce the real-coefficient polynomial q_0 by the exact identity $\tilde{Q}_0 = 2iq_0$. Thus its four-factor conductor has the scalar $(2i)^4$ as well. In the order $(Y_1, Y_3; Y_2, Y_4)$, write

$$q_0(Y) = (Y_1, Y_3)A(Y_1, Y_3)^T + (Y_2, Y_4)B(Y_2, Y_4)^T. \quad (\text{NC13})$$

The real symmetric blocks are specified without any unit-phase choice:

$$A = \begin{pmatrix} \Im c & \Im(c(-t + id)) \\ \Im(c(-t + id)) & \Im(c(-t + id)^2) \end{pmatrix}, \quad B = - \begin{pmatrix} \Im(c\alpha) & \Im(c\alpha(t + id)) \\ \Im(c\alpha(t + id)) & \Im(c\alpha(t + id)^2) \end{pmatrix}. \quad (\text{NC14})$$

For complex C and L , direct expansion yields

$$\Im C \Im(CL^2) - \Im(CL)^2 = -|C|^2(\Im L)^2.$$

It follows that

$$\det A = -|c|^2 d^2 < 0, \quad \det B = -|c|^2 |\alpha|^2 d^2 < 0. \quad (\text{NC15})$$

In particular both real binary quadratic forms are nonsingular and indefinite. No special unit value has been imposed.

4 All coordinate-plane intersections at the original limit

Lemma 2. *If $0 < d \leq t$, no diagonal entry of A and diagonal entry of B can both be zero. Neither pair of diagonal entries of a single block can both be zero.*

Proof. The four diagonal coefficients are $\Im(cf_j)$ for

$$f_1 = 1, \quad f_2 = (-t + id)^2, \quad f_3 = -\alpha, \quad f_4 = -\alpha(t + id)^2.$$

If $\Im(cf_j) = \Im(cf_l) = 0$, then $f_j \bar{f}_l$ is real. The exact cross-block values are

$\Im(f_j \bar{f}_l)$	$l = 3$	$l = 4$	(NC16)
$j = 1$	d	$-d(d^2 + 5t^2)$	
$j = 2$	$-d(d^2 + 5t^2)$	$d(d^4 + 6d^2t^2 - 11t^4)$	

The bottom right entry is negative for $0 < d \leq t$, since its parenthesis is at most $-4t^4$. The other entries are nonzero. Within a block, the ratios $f_2/f_1 = (-t + id)^2$ and $f_4/f_3 = (t + id)^2$ have nonzero imaginary parts. These observations prove both assertions, for every nonzero c . $\square$

Theorem 3. *If $0 < d \leq t$, then for every $i = 1, 2, 3, 4$ and $j = 1, 2, 3, 4$ the two ternary quadrics*

$$q_0|_{Y_i=0}, \quad q_0(D^{-j}Y)|_{Y_i=0} \tag{NC17}$$

are coprime. Therefore the limiting conductor is coprime to the product of the four coordinate sections on the original quadric.

Proof. Deleting one variable from a block in (NC13) leaves a single square in that block and the other nonsingular binary form. If the square coefficient is nonzero, the ternary form has rank three, hence is an irreducible conic over $\mathbb{C}$. A common factor with its diagonal transform would then force the two conics to be proportional. For $j \not\equiv 0 \pmod{5}$, proportionality of a polynomial and its transform requires every nonzero monomial to have the same weight modulo five. The weights of the six possible monomials are

monomial	Y_1^2	Y_1Y_3	Y_3^2	Y_2^2	Y_2Y_4	Y_4^2	(NC18)
weight	2	4	1	4	1	3	

On $Y_2 = 0$ or $Y_3 = 0$ all remaining monomial weights are distinct; rank three excludes a one-monomial polynomial. On $Y_1 = 0$, the only weight supporting two monomials is weight one, requiring both diagonal entries of B to vanish. On $Y_4 = 0$, it is weight four, requiring both diagonal entries of A to vanish. The preceding lemma excludes both. Thus the irreducible case is settled.

If the square coefficient is zero, the ternary conic is the other binary form. That form has two distinct real projective roots by (NC15). Both are finite and nonzero: a zero or infinite root would make a diagonal entry of that block zero, which (NC16) forbids simultaneously with the deleted block's diagonal coefficient. The ratio of its two coordinates is multiplied under D^{-j} by ζ^{2j} or its inverse. For $1 \leq j \leq 4$ this number is nonreal. Multiplying a finite nonzero real root by it cannot yield either real root. Hence the two reducible conics share no line. These cases exhaust every coordinate-plane section and every retained unit phase.

A common divisor of a coordinate section and the conductor on the Segre surface would give a curve on $Q_0 = Q_j = Y_i = 0$ for some i, j . In the plane $Y_i = 0$ this is precisely a common polynomial factor in (NC17), just excluded. This proves the last assertion. $\square$

The condition is established for every actual hypothetical off-line quartet: $\gamma \geq 7\delta$ gives $d/t = 6\delta\gamma/(\gamma^2 - \delta^2) \leq 7/8 < 1$. Platt and Trudgian's rigorously verified finite-height theorem [3] places any such zero above $3 \cdot 10^{12}$, whereas $\delta < 1/2$. Only the much weaker consequence $\gamma \geq 7\delta$ is used here. The cited theorem is a human analytic input; the coprimality argument above is proved independently for its full stated parameter domain. For a formal non-zeta quartet outside that domain, the finite matrices below still decide coprimality, but the preceding universal nonvanishing argument is not asserted there.

5 Sixteen finite matrices give an original finite-period bound

For each i, j let $q_i(z)$ and $q_{ij}(z)$ be the restrictions to $Y_i = 0$ of $\tilde{Q}_z(Y)$ and $\tilde{Q}_z(D^{-j}Y)$. They are ternary quadrics with entire coefficient functions. In the unchanged ordered monomial bases define the 10 by 6 matrix

$$M_{ij}(z) : (L_1, L_2) \mapsto L_1 q_i(z) + L_2 q_{ij}(z), \quad (L_1, L_2) \in T_1(\mathbb{C}^3)^2, \quad M_{ij}(z)(L_1, L_2) \in T_3(\mathbb{C}^3). \quad (\text{NC19})$$

Here $T_e(\mathbb{C}^3)$ means the homogeneous ternary forms of degree e ; it is not the biform notation used earlier. Neither restricted quadric is zero, since an invertible transform of Q_0 cannot have a coordinate hyperplane as a factor.

Lemma 4. *The matrix $M_{ij}(z)$ has rank six if and only if its two quadrics are coprime.*

Proof. If they are coprime, $L_1 q_i = -L_2 q_{ij}$ implies $q_i \mid L_2$ in the polynomial UFD. Degree forces $L_2 = 0$, then $L_1 = 0$. If their gcd has degree one, write them as ga, gb with a, b linear. The pair $(b, -a)$ is a nonzero kernel vector. If the gcd has degree two, the quadrics are proportional and multiplication of that constant relation by any linear form gives a kernel vector. $\square$

Expand $M_{ij}(z) = \sum_{n \geq 0} M_{ij,n} z^n$, using exactly (NC4), the polynomial adjugate $\mathcal{R}^{-1} = \text{adj } \mathcal{R}$, and (NC9). In Euclidean coefficient norms set

$$\mu_{ij} = \sigma_{\min}(M_{ij,0}) > 0, \quad C_{ij} = \sum_{n \geq 1} \|M_{ij,n}\|_2 < \infty, \quad R_{\text{cop}} = \max \left(1, \max_{i,j} \frac{2C_{ij}}{\mu_{ij}} \right). \quad (\text{NC20})$$

The just-proved rank theorem establishes each strict positivity; it is not an assumed nonzero minor. Entire convergence gives the summability in (NC20), for example by a Cauchy bound on the circle of radius two. All coefficients in this finite list of specified series are given by the original packet, its actual unit and (NC4). For $|u| \geq R_{\text{cop}}$,

$$\|M_{ij}(u^{-1}) - M_{ij,0}\|_2 \leq C_{ij}/|u| \leq \mu_{ij}/2, \quad \sigma_{\min}(M_{ij}(u^{-1})) \geq \mu_{ij}/2 > 0. \quad (\text{NC21})$$

This proves coprimality at finite original periods. If $C_{ij} = 0$ its matrix is constant and the same conclusion follows immediately.

One can also retain the whole original domain $|u| \geq R_* > 0$. For each matrix choose the first ordered nonzero 6 by 6 minor at $z = 0$, and let $f_{ij}(z)$ denote that same minor at z . Their product $F(z)$ is entire and $F(0) \neq 0$. Its zeros in $|z| \leq R_*^{-1}$ are finite. Outside that explicitly specified finite set every matrix has rank six. At a zero of F , use all minors of (NC19): a zero of the selected minor alone does not assert failure.

Theorem 5. *For an actual hypothetical simple off-line quartet, on the original five-orbit domain and every fixed period $|u| \geq R_{\text{cop}}$,*

$$\gcd(\mathcal{A}, (\ell_1 \ell_2 \ell_3 \ell_4)^4) = 1, \quad E_k = 0 \quad (k \geq 9). \quad (\text{NC22})$$

The same conclusion holds at every admitted period outside the finite exceptional set specified above. Period, branch and unit are fixed while k grows.

Proof. Apply (NC21), the rank criterion, the coordinate-plane argument, the exact pullback (NC9), and (NC8), in that order. A divisor occurring with higher multiplicity would in particular occur once, already excluded. This retains every factor multiplicity of the original conductor. The admissible five-orbit radius, if larger, is imposed by taking the maximum with R_{cop} . $\square$

6 Return to the original rank and attained metric

The supplied low-jet calculation J1–6 [1] defines $v = \text{ord}_0 E_A$ at the same actual period. On its two-active-coordinate locus it proves $t_0(k, u) = 0$ for $k \geq 5v - 1$. It also proves that the one-coordinate exceptions in the admitted inverse-period disk lie in a finite set. Intersecting this locus with (NC22) therefore gives

$$j_{0,k} = 8k - 32 - v, \quad r_I = j_{0,k} - \dim E_k = 8k - 32 - v \quad (k \geq \max(9, 5v - 1)). \quad (\text{NC23})$$

At all other periods the exact identity is still $r_I = j_{0,k} - \dim E_k$ with the actual finite jet rank and conductor gcd; the displayed value of t_0 is not substituted there.

Here is the actual metric consequence. Use the original notation of the proper-source map: Z_R is its ordered value-space inclusion, J_0 its coefficient lift, and $D_{k,N-v}^{(s_0)}$ the original degree- k , source-order- s_0 attained form at its original cutoff. Before (NC22), this source metric is the attained quotient by E_k : for an E_k basis matrix Z_E it is

$$Q_N^p = Z_R^* J_0^* D J_0 Z_R - Z_R^* J_0^* D J_0 Z_E (Z_E^* J_0^* D J_0 Z_E)^{-1} Z_E^* J_0^* D J_0 Z_R, \quad D = D_{k,N-v}^{(s_0)}. \quad (\text{NC24})$$

This is obtained by completing the square in the actual affine fiber; the second term uses the full cross block. On (NC22) the fiber direction space is zero. Consequently the exact receiving equality is

$$Q_N^p = Z_R^* J_0^* D_{k,N-v}^{(s_0)} J_0 Z_R. \quad (\text{NC25})$$

The target metric retains its entire target direction space $\mathcal{J}$. If W is its actual value lift, T a basis of $\mathcal{J}$, and G the original target form, completing the square still gives

$$H_N^p = W^* G W - W^* G T (T^* G T)^{-1} T^* G W. \quad (\text{NC26})$$

No direction in $\mathcal{J}$ has been removed by (NC22). The original relative determinant return is thus exactly

$$\mathcal{M}^p = \mathcal{R}_{\text{end}} \log \det \left[(Z_R^* J_0^* D J_0 Z_R)^{-1/2} H_N^p (Z_R^* J_0^* D J_0 Z_R)^{-1/2} \right]. \quad (\text{NC27})$$

$\mathcal{R}_{\text{end}}$ applies the original four endpoint signs. Equation (NC27) removes an actual source minimization. Evaluation of its remaining target minimum and native relative spectrum requires the original matrices in (NC26); the ambient displacement coefficient in D24 is not inserted as their determinant.

7 Backward receiving statements and the next calculation

The following replacements are precise consequences for the supplied four notes. In E4, the coefficient $(g_z + g_w)(k - 7) + g_z g_w$ is zero on (NC22), and E_k itself is zero. In its final metric paragraph, the previously unevaluated coprime stratum is now proved to contain the finite-period domain (NC20)–(NC21), for every actual retained unit phase. J6 keeps the original observation kernel and is unchanged; (NC23) concerns the independent proper-source rank. O9–12 continue to recover the original full packet by observation iterates; none of their inverses has been substituted into the target minimum (NC26).

The scalar results D19, D20, D23 and D24 retain their original four-endpoint domains and analytic providers. The two supplied auxiliary scripts check their displayed coefficient identities and character counts (11 and 45 checks); those checks do not constitute another proof of the imported analytic estimates. The four complete incoming proof notes accompany this continuation. The separately supplied low-jet note adds the smaller one-coordinate matrix J3a and explicit classes J3b and is retained as the newer version.

The next numerical or asymptotic metric calculation now starts with the literal positive pair in (NC25)–(NC26), at the proper-source displaced cutoffs, with no source E_k block. This is

the map on which the remaining relative eigenvalues and determinant must be computed. The theorem does not assign individual projected signs or the seven absolute allocations from this rank result.

References

- [1] *Programme Native Continuation*, supplied 19 September 2026: “Displaced lower covariance and matched native return through order q ” (D1–25); “Original observation: cofinal low-degree kernel ranks” (J1–6, including J3a–b in the separately supplied revision); “The original proper-source kernel is annihilated on the coordinate divisor” (E1–4); “Full original packet recovery from iterated observation” (O1–12). Complete source files are included with this continuation, with their original bytes preserved.
- [2] *The literal original conductor limit and an invertible pivot restriction*, PCL1–7 and PCL2’s complete coefficient series, existing programme proof. Included as `PCL_COMPLETE.tex`. The scalar, diagonal, branch, Vandermonde and arithmetic unit factors are those of PCL7.
- [3] Dave Platt and Tim Trudgian, *The Riemann hypothesis is true up to $3 \cdot 10^{12}$* , Bulletin of the London Mathematical Society **53** (2021), 792–797. doi:10.1112/blms.12460; arXiv:2004.09765. Used only for the proved finite-height exclusion of off-line zeros in the parameter-domain check following (NC18).